Sarthak Khare

Aspiring Machine Learning Engineer | Cybersecurity Enthusiast

+91-8269720041 | kharesarthak86@gmail.com

Profile

Computer Science student specializing in Data Science with hands-on experience in Machine Learning, Cybersecurity and Deep Learning. Skilled in building intelligent systems including Intrusion Detection Systems and Deepfake Detection models using Python and TensorFlow. Passionate about solving real-world problems through AI-driven secure solutions.

Education

Manipal University Jaipur — Jaipur (Rajasthan)
Bachelor of Technology in Computer Science & Engineering Data Science (2023-27) — **CGPA – 8.02**

St. Joseph's Convent Sr. Sec School Ranjhi — Jabalpur (M.P.)
Higher Secondary (2022) — **Percentage – 80.2%**
Secondary (2020) — **Percentage – 85.8%**

Projects

Web Crawler | *Python, React, FastAPI, Selenium, Sentence-BERT, DistilBERT, OpenCV, Celery, Redis*

Developed an AI-powered web crawler that detects 9 types of deceptive UI tricks in websites using NLP, ML classification and Computer Vision. Implemented sematic focused crawling with Sentence-BERT embeddings, hybrid detection combining rule-based matching with DistilBERT sentiment analysis, and OpenCV screenshot analysis for visual dark patterns. Built a React dashboard with live progress tracking, interactive site graph and pattern alerts with confidence scoring.

Intrusion Detection System | *Python, Pandas, NumPy, Scikit-learn, XGBoost, MLPClassifier*

Developed a machine learning based Intrusion detection System (IDS) to classify IOT network traffic from Philips as a normal or malicious based on the protocol and behavioral patterns. Performed data cleaning, label encoding, and feature engineering to extract meaningful attributes from packet data. Applied supervised learning algorithms including Random Forest, SVM, Logistic Regression and Neural Networks for network anomaly classification.

DeepFake Video and Image Detection System | *TensorFlow, Keras, Python, EfficientNetB4, NumPy, OpenCV*

Developed a machine learning based Deepfake Detection System (DFDS) using **Transfer Learning** with **EfficientNetB4** CNN architecture. The system classifies images as 'Real' or 'Fake' by incorporating robust **Data Augmentation** and strategic callbacks (**ModelCheckpoints/ EarlyStopping**) to prevent overfitting, resulting in a highly accurate and efficient binary classification solution.

Other Projects – Ticket booking System using Java, Snake Game utilizing data structures and algorithm in Java.

Technical Skills

Programming Languages: Python, Java, C, SQL
Machine Learning: Supervised Learning, Classification, Feature Engineering, Data Preprocessing, Model Evaluation
Frameworks & Libraries: Scikit-learn, TensorFlow, Keras, OpenCV, Pandas, NumPy, XGBoost, MLP Classifier
Tools: Git, VS Code, Anaconda, Linux, Wireshark, N-Map
Domains: Cybersecurity, Intrusion Detection Systems, Computer Vision, Software Testing
Courseworks: Data Structures and Algorithms, Operating Systems, OOPs, RDBMS, Computer Networks

Achievements

- **IMUN :-** Participated in a debate organized by MUN and received an appreciation letter via IMUN 194.0.
- **TECH TORNADO 2.0 :-** Secured 2nd place at TECH TORNADO 2.0 conducted by Cypher Club at Manipal University Jaipur.
- **SIH 2025 :-** Secured top 10 position at internal round in Smart India Hackathon conducted by Manipal University Jaipur.

Certifications

Foundations of Cybersecurity	Google
Play It Safe: Manage Security Risks	Google
Connect and Protect: Network and Network Security	Google
Machine Learning Specialization	Stanford Online | DeepLearning.AI
Cryptography	(ISC)^2